

Multiple Fusion Feature Combination Model for Enhanced Multimodal Sentiment Analysis

Liang Yang, Xinhua Zhu

Abstract—Multimodal sentiment analysis (MSA) aims to determine sentiment polarity by integrating and fusing text, audio, and visual cues. However, insufficient modality interaction is an ongoing problem in the existing methods. This paper proposes a multiple fusion feature combination model for MSA. First, we design a serial cross-modal interaction method that sequentially integrates text, audio, and visual modalities through cross-modal attention mechanisms, enhanced with gated recurrent unit networks to filter the redundant information. We then design a single-mode semantic enhancement module that fully mines the local and global semantics to obtain an enhanced fusion feature, and introduce a weighted fusion approach that elevates the semantic contributions of visual and audio modalities during the fusion process. Finally, we combine the three fused features to predict the user’s final emotional state, utilizing the mean absolute error and cosine similarity in the loss calculation to improve the prediction accuracy. When evaluated on the CMU-MOSI and CMU-MOSEI datasets, our model outperformed several state-of-the-art approaches, showcasing its ability to enhance MSA performance.

Index Terms—Multimodal sentiment analysis, Deep learning, modal interaction.

I. INTRODUCTION

Sentiment analysis, which is used in personalized advertising, opinion mining[1], and decision-making, has evolved from dictionary-based[2][3] methods to deep learning[4]. Short video platforms have promoted the expression of human emotions through videos, providing abundant multimodal data. Traditional sentiment analysis focuses on text, but human emotions are multidimensional[5][6], involving voice, facial expressions, and body language. Thus, sentiment analysis is evolving toward multimodal sentiment analysis (MSA), which processes multiple modal information for a comprehensive emotion understanding of emotions. MSA enables machines to

accurately recognize emotions across various applications[7], but requires the effective fusion of modality-specific features. The existing fusion methods include early, late, and hybrid fusion. Direct operations used in early research, such as concatenation and weighted sum, cannot adequately capture the rich semantic interactions. Attention mechanisms[8] improve this situation by exploring the fine-grained interactions between modalities. However, most attention-based[9][10] models use parallel structures for pairwise interactions, ignoring the deeper serial fusion that might reveal deeper correlations. Current MSA models primarily focus on eliminating modality differences rather than exploring diverse fusion methods.

This paper proposes a multiple fusion feature combination model for MSA, aiming at feature fusion and association mining through three fusion methods while enhancing the connection and complementarity of these methods. To enhance the parallel structure of cross-modality multihead attention models, we propose a serial cross-modal interaction method using a two-layer network with multihead attention and gated recurrent units (GRUs) [11]. The first layer utilizes text as queries and audio as keys and values. The interaction result of the first layer becomes the query in the second layer, which interacts with visual data. We then propose a semantic enhancement module that enhances modalities from multiple levels. Visual and audio features are often considered as low-level semantic features, containing large amounts of redundant information. Therefore, we also designed a cross-modality alignment module that elevates the semantic alignment of visual and audio modalities to that of text to improve the fusion effects. Our contributions are summarized below.

- 1) We propose a multiple fusion feature combination model that fuses the three modal features (text, audio, and visual) using three fusion methods. This model aims to enhance the complementarity of the various fusion methods.
- 2) We propose a serial cross-modal interaction method using a two-layer network of multihead attention and GRU. This method improves the parallel structure of cross-modality multihead attention models, thereby enhancing their modality connections and avoiding redundancy.
- 3) We design a semantic enhancement module employing one-dimensional convolutional networks and attention mechanisms to locally and globally enhance the semantic representation of modal features.
- 4) We propose a weighted fusion method that elevates the semantic information of visual and audio to the level of text for

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fusion, enhancing the contribution of semantic information to the model.

II. APPROACH

The overall structure of our model is shown in Fig. 2.

A. Feature Extraction

The pretrained Bidirectional Encoder Representation from Transformers (BERT)[12] model excellently extracts the text features using a bidirectional transformer encoder and learns

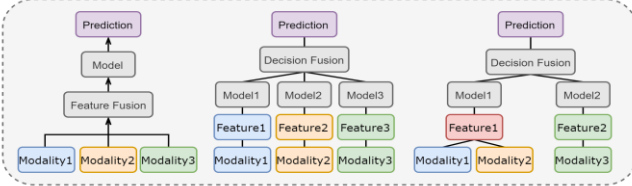


Fig. 1 Three widely used fusion methods.

the context information through large-scale unsupervised pretraining. In our approach, BERT extracts the text information from an input text sequence X_t , taking the first word vector of the last layer as the feature representation, and then transforms it to a unified dimension through a linear layer:

$$f_t = \text{Linear}(BERT(X_t; \theta_t^{bert})) \in \mathbb{R}^{d_t}, \quad (1)$$

where θ_t^{bert} denotes all learnable parameters of BERT, and d_t denotes a custom unified dimension for the text feature.

For the audio and vision modalities, we extract the initial vector features $X_a \in \mathbb{R}^{l_a \times d}$ and $X_v \in \mathbb{R}^{l_v \times d}$, respectively, using pretrained Tool-Kits[13]. Here l_a and l_v are the lengths of the audio and visual sequences, respectively. Next, the timing characteristics are captured by single-directional stacked long short-term memory (LSTM), which alleviates the gradient issues in traditional recurrent neural networks. Finally, these features are transformed to a unified dimension through a linear layer:

$$f_a = \text{Linear}(sLSTM(X_a; \theta_a^{sLSTM})) \in \mathbb{R}^{d_a} \quad (2)$$

$$f_v = \text{Linear}(sLSTM(X_v; \theta_v^{sLSTM})) \in \mathbb{R}^{d_v} \quad (3)$$

where θ_a^{sLSTM} and θ_v^{sLSTM} denote all learnable parameters of the LSTM, and d_a and d_v denote the custom unified dimensions of the audio and visual sequences, respectively.

B. Serial Cross-modality Attention Fusion Module

This module uses text as the main modality and sequentially interacts with the audio and visual modalities to obtain cross-modal features. Initially, we use the text feature f_t as the query and the audio feature f_a as the key and value to calculate an audio-enhanced text feature. After filtering through a GRU network, we obtain F_t^{a-enh} . Taking F_t^{a-enh} as the query and the visual feature f_v as the key and value, we then repeat the operation to obtain $F_t^{a,v-enh}$. This module explores the semantic information similar to text in audio and visual modalities, filtering out redundant information to enhance the fusion semantic feature. Define $Q_t = W^t \cdot f_t \in \mathbb{R}^{d_t}$ and $K_a = V_a = W^a \cdot f_a \in \mathbb{R}^{d_a}$, the cross-modal attention is computed as follows:

$$\text{CrossAttn}(Q_t, K_a, V_a) = \text{softmax}\left(\frac{Q_t \cdot (K_a)^T}{\sqrt{d_k}}\right) \cdot V_a \quad (4)$$

For the i th head in cross-modal multihead attention, the calculation becomes

$$\hat{Q}_{ta,i} = \text{CrossAttn}(W_i^t \cdot Q_{t,i}, W_i^a \cdot K_{a,i}, W_i^a \cdot V_{a,i}) \in \mathbb{R}^{\frac{d_t}{n}} \quad (5)$$

where W_i^t and W_i^a are the projection matrices for the i th head, n denotes the head number, and $\hat{Q}_{ta} \in \mathbb{R}^{d_t}$ is the audio-enhanced text feature, computed as follows:

$$\hat{Q}_{ta} = \text{CrossMHA}(Q_t, K_a, V_a) = \text{concat}(\hat{Q}_{ta,1} \dots \hat{Q}_{ta,n}) \cdot W^o \quad (6)$$

To prohibit information unrelated to the text features during the cross-fusion process, we partially filter the noise through a GRU network:

$$F_t^{a-enh} = \text{GRU}(\hat{Q}_{ta}; \theta^{GRU}) \in \mathbb{R}^{d_t} \quad (7)$$

Subsequently, the visual feature f_v is fused through identical procedures. Define $Q_{ta} = W^{ta} \cdot F_t^{a-enh} \in \mathbb{R}^{d_t}$ and $K_v = V_v =$

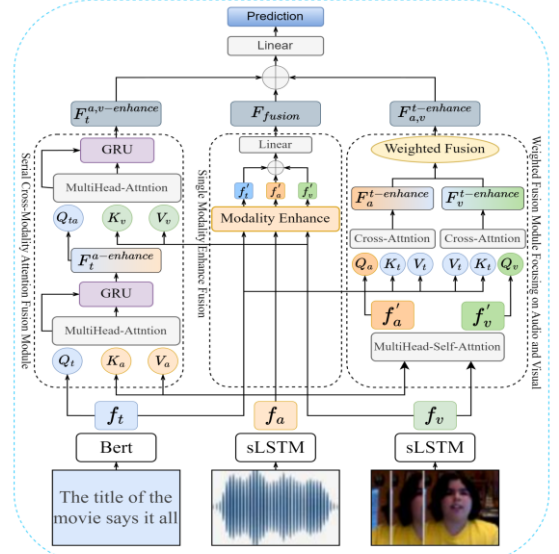


Fig. 2. Overall architecture of the proposed model.

$W^v \cdot f_v \in \mathbb{R}^{d_v}$, $F_t^{a,v-enh}$ is computed as follows:

$$\hat{Q}_{tav,i} = \text{CrossAttn}(W_i^{ta} \cdot Q_{ta,i}, W_i^v \cdot K_{v,i}, W_i^v \cdot V_{v,i}) \in \mathbb{R}^{\frac{d_t}{n}} \quad (8)$$

$$\begin{aligned} \hat{Q}_{tav} &= \text{CrossMHA}(Q_{ta}, K_v, V_v) \\ &= \text{concat}(\hat{Q}_{tav,1} \dots \hat{Q}_{tav,n}) \cdot W^o \end{aligned} \quad (9)$$

$$F_t^{a,v-enh} = \text{Linear}(\text{GRU}(\hat{Q}_{tav}; \theta^{GRU})) \in \mathbb{R}^{d_f} \quad (10)$$

where d_f denotes the custom fusion feature dimension, W^{ta} , W^v and W^o are weight matrices, and θ^{GRU} denotes all learnable parameters of GRU.

C. Single Modality Enhancement Fusion

This section discusses our second fusion method, namely, single-modal enhancement fusion. Utilizing semantic enhancement techniques, this method thoroughly explores the rich semantic information embedded in each modality, as shown in Fig. 3. First, the model produces three copies of the three modal features: one unchanged replicate, one copy to be processed by a convolutional network for local semantic extraction, and one copy to be processed by a multihead attention mechanism for long-range dependencies. The first and second copies are summed and passed through a linear layer. Repeating this process with the fused features and the third copy, it obtains the enhanced modal features as follows:

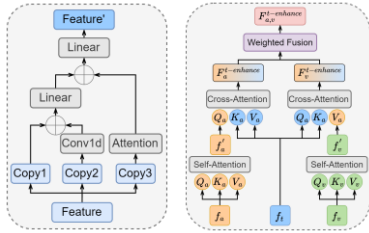


Fig. 3. Semantic enhancement module (left) and weighted fusion of visual and audio features (right).

$$F_m^{local} = Linear([F_m^{copy1}; Conv1d(F_m^{copy2}; K_m)]) \quad (11)$$

$$F_{fusion} = Linear([F_m^{local}; Attn(F_m^{copy3}, F_m^{copy3}, F_m^{copy3})]) \quad (12)$$

where k_m denotes the kernel size of each modality, $m \in \{t, v, a\}$.

D. Weighted fusion of the visual and audio features

To enhance the semantic granularity and align the visual and audio modalities with the text, we designed a weighted fusion module that processes the audio feature f_a and visual features f_v using self-attention. The resulting concentrated features f'_a and f'_v , respectively, serve as queries in the cross-modal attention mechanism, which uses the text feature f_t as the key and value to produce enhanced features $f_a^{t-enhance}$ and $f_v^{t-enhance}$. After weighted concatenation, the module yields the fusion feature $f_{a,v}^{t-enhance}$. By dynamically adjusting the attention to text based on the audio and visual features, this module improves the extraction accuracy and promotes the fusion.

Defining $Q_a = K_a = V_a = W^a \cdot f_a \in \mathbb{R}^{d_a}$, f'_a is computed as follows:

$$f'_a = SelfAttention(Q_a, K_a, V_a) \in \mathbb{R}^{d_a} \quad (13)$$

Next, defining $Q_a = W'_a \cdot f'_a \in \mathbb{R}^{d_a}$ and $K_t = V_t = W^t \cdot f_t \in \mathbb{R}^{d_t}$, the enhanced audio feature F_a^{t-enh} is obtained by the cross-attention module as:

$$F_a^{t-enh} = CrossAttn(Q_a, K_t, V_t) \in \mathbb{R}^{d_a} \quad (14)$$

Repeating this process, we obtain the enhanced visual feature F_v^{t-enh} . Finally, after weighting and concatenating F_a^{t-enh} and F_v^{t-enh} , we obtain the weighted fusion feature $F_{a,v}^{t-enh}$:

$$F_{a,v}^{t-enh} = Linear([\alpha \cdot F_a^{t-enh}; \beta \cdot F_v^{t-enh}]) \in \mathbb{R}^{d_f} \quad (15)$$

where α and β are the weight parameters of F_a^{t-enh} and F_v^{t-enh} , respectively.

Finally, we sum the fusion vectors $F_t^{a,v-enh}$, F_{fusion} and $F_{a,v}^{t-enh}$, and pass the result through the linear layer. The resulting fusion representation Y^p is used in the sentiment prediction process:

$$Y^p = W_2^T (Relu(W_1^T [F_t^{a,v-enh}; F_{fusion}; F_{a,v}^{t-enh}] + b_1)) + b_2 \quad (16)$$

where $W_1^T \in \mathbb{R}^{3d_f \times d_f}$, and $W_2^T \in \mathbb{R}^{d_f \times 1}$.

E. Objective Function

The average distance between the predicted value Y^p and the true value $Y^t \in [-3, 3]$ is determined using the mean absolute error:

$$\mathcal{L}_{mae} = \frac{1}{n} \sum_{i=1}^n |Y_i^t - Y_i^p| \quad (17)$$

where n denotes the batch size. In addition, we calculated the cosine similarity between the predicted positive values and positive labels and between the predicted negative values and negative labels. The weighted sum of these similarities improves the binary classification accuracy. The calculation is as follows:

$$\mathcal{L}_{cos} = 1 - (\lambda_p \cdot cosine(Y^{p+}, Y^{t+}) + \lambda_n \cdot cosine(Y^{p-}, Y^{t-})) \quad (18)$$

where $cosine(\cdot)$ represents the cosine similarity, and λ_p and λ_n represent the positive and negative weights, respectively. Finally, our loss function is designed as follows:

$$\mathcal{L}_{task} = \lambda_1 \cdot \mathcal{L}_{mae} + \lambda_2 \cdot \mathcal{L}_{cos} \quad (19)$$

III. EXPERIMENTAL RESULT

A. Implementation Details

Training Details: The proposed model was implemented in Python 3.9 and Pytorch 1.4.0 running on a Windows 10 system with a single NVIDIA Tesla V100 (32G) GPU. The pretrained BERT model was sourced from Hugging Face's Transformer toolkit. The hyper-parameter settings during the training process are shown in Table. 1.

Table. 1. Experimental parameters.

Hyper-Parameter	Value
Optimizer	Adam
Batch Size	32
Learning rate on text	(5e-5)
Learning rate on audio and visual	(1e-5)
hidden size d_t , d_v , and d_a	128
fusion size d_f	128

Datasets: Experiments were performed on two publicly available multimodal sentiment analysis datasets: CMU-MOSI[14] and CMU-MOSEI[15]. The basic information of the two datasets is shown in Table. 2.

Table. 2. Datasets statistics of CMU-MOSI and CMU-MOSEI.

Datasets	Train	Valid	Test	Total
CMU-MOSI	1283	229	686	2198
CMU-MOSEI	16326	1867	4659	22852

Comparison Methods: The proposed model was competed against four classical models (ICCN[16], MuT[17], MISA[18], and Self-MM[19]) and four recent state-of-the-art models (DMD[20], ALMT[21], TETFN[22], and TMBL[23]).

B. Comparisons with State-of-the-Art Methods

The experimental results are shown in Table. 3(MOSI) and Table. 4(MOSEI).

1) *Quantitative Comparisons:* In experiments on the CMU-MOSI dataset, our model achieved an MAE of 0.712, a Corr of 0.803, an Acc-2 of 86.28, and an F1 score of 86.29, outperforming all comparison models. It also secured the second-best Acc-7 score (46.43), trailing only ALMT. On the CMU-MOSEI dataset, our model outperformed the baseline models across all metrics, achieving SOTA performance.

Table. 3. Result on CMU-MOSI.

Models	MAE(↓)	Corr	Acc-2	F-score	Acc-7
ICCN	0.862	0.714	−83.0	−83.0	39.0
MuT	0.861	0.711	81.5/84.1	80.6/83.9	40.0
MISA	0.783	0.761	81.8/83.4	81.7/83.6	—
Self-MM	0.713	0.798	84.0/85.98	84.42/85.95	—
TETFN	0.717	0.800	84.05/86.10	83.83/86.07	—
TMBL	0.867	0.762	81.78/83.84	82.41/84.29	36.3
DMD*	0.736	0.784	85.4	85.4	46.34
ALMT*	0.720	0.783	82.94/84.91	83.07/84.97	47.08
Ours	0.707	0.803	84.69/86.59	84.65/86.59	46.43

Table. 4. Result on CMU-MOSEI.

Models	MAE(↓)	Corr	Acc-2	F-score	Acc-7
ICCN	0.565	0.713	−84.2	−84.2	51.6
MuT	0.580	0.703	−82.5	−82.3	—
MISA	0.568	0.724	82.59/84.23	82.67/83.97	—
Self-MM	0.530	0.765	82.81/85.17	82.53/85.30	—
TETFN	0.551	0.748	84.25/85.18	84.18/85.27	52.5
TMBL	0.545	0.766	84.23/85.84	84.87/85.92	52.4
DMD*	0.541	0.762	85.5	85.4	52.5
ALMT*	0.533	0.768	82.23/85.66	81.89/85.79	52.82
Ours	0.527	0.769	83.13/86.3	83.5/86.24	54.8

2) *Qualitative Comparisons:* As shown in Fig. 4, the loss metric largely decreased with increasing number of training epochs, indicating that the model performance improves with training. The smoothed loss curve in later stages suggests that the stability also improve. Visualization of t-distributed stochastic neighbor embedding initially shows chaotic fused representations. As the training progressed, the representations became clustered by emotional polarity.

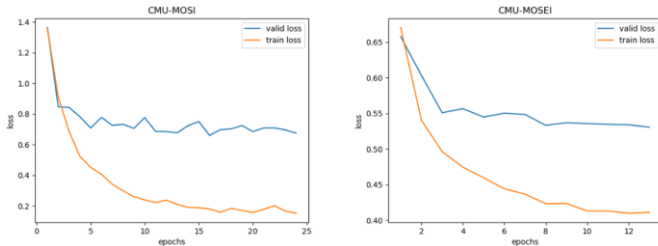


Fig. 4. Losses of our method on the CMU-MOSI (left) and CMU-MOSEI (right) datasets.

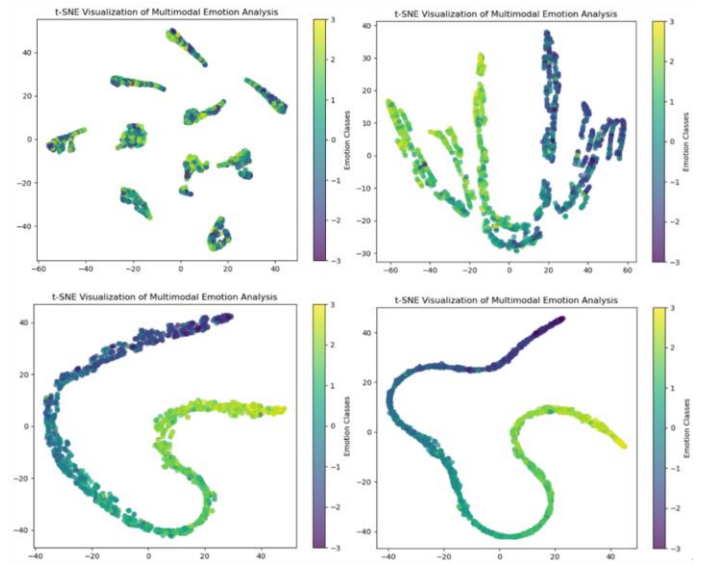


Fig. 5. T-SNE visualization during the fusion process.

3) Ablation study

The ablation experiments separately evaluated the effectiveness of the three fusion methods and their pairwise combinations. Omitting one or both methods was shown to degrade the performance of our model.

The final performance of various fusion combinations is detailed in Table. 5 (CMU-MOSI) and Table. 6 (CMU-MOSEI).

Table. 5. The results of various fusion methods on CMU-MOSI.

Modality	MAE	Corr	Acc-2	F-score	Acc-7
F1	0.726	0.790	83.11/84.2	82.93/84.34	45.96
F2	0.728	0.782	82.8/84.6	82.73/84.59	45.91
F3	0.743	0.779	82.65/84.45	82.63/84.48	43.29
F1+F2	0.735	0.788	83.24/85.21	83.2/85.23	46.11
F1+F3	0.721	0.791	82.22/83.54	82.19/83.56	45.34
F2+F3	0.744	0.792	82.65/84.3	82.66/84.35	44.75
F1+F2+F3	0.707	0.803	84.55/86.28	84.51/86.29	46.21
Promote	0.14	0.011	1.31/1.07	1.31/1.06	0.10

Table. 6. The results of various fusion methods on CMU-MOSEI.

Modality	MAE	Corr	Acc-2	F-score	Acc-7
F1	0.538	0.759	82.94/84.84	82.97/84.84	54.09
F2	0.546	0.755	82.64/84.44	82.77/84.24	53.09
F3	0.545	0.757	82.4/84.81	82.53/84.52	53.08
F1+F2	0.537	0.760	82.66/84.59	82.74/84.29	54.22
F1+F3	0.541	0.762	82.64/84.2	82.83/84.98	53.86
F2+F3	0.536	0.755	82.24/84.59	82.25/84.28	54.22
F1+F2+F3	0.527	0.769	83.13/86.3	83.5/86.24	54.8
Promote	0.08	0.007	0.88/0.98	0.78/0.83	0.58

IV. CONCLUSION

This paper introduced a multimodal sentiment analysis model that integrates early and model-level fusion, leveraging multitype and multilevel fusion to enhance intermodal connections. When evaluated on the CMU-MOSI and CMU-MOSEI datasets, our model outperformed several existing methods. Empirical results indicated that the textual features often surpass the nontextual ones. In future work, we will investigate the roles of the strong and weak features in MSA.

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